6.1 Inner Product, Length & Orthogonality

In order to solve problems like line of best fit, Fourier approximations and more, we must develop fundamental ideas of *length*, *orthogonality* and *orthogonal projections*.

The Inner Product. In R^2, we called the inner product the dot product.

Inner product or dot product of $u = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ $\quad u \bullet v = u^T v = \begin{bmatrix} u_1 & u_2 & \dots & u_n \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix} = u_1 v_1 + u_2 v_2 + \cdots + u_n v_n$

Note that $v \bullet u = v_1 u_1 + \cdots + v_n u_n = u_1 v_1 + \cdots + u_n v_n = u \bullet v =$

Ex 1: Compute each of the following given that $u = \begin{bmatrix} 2 \\ -1 \\ 3 \\ 2 \end{bmatrix}, v = \begin{bmatrix} -1 \\ 0 \\ 2 \\ 5 \end{bmatrix}$ *and* $w = \begin{bmatrix} 4 \\ -3 \\ 0 \\ 6 \end{bmatrix}$

a. $v \bullet w$ b. $u \bullet (v - w)$ c. $\| u \|$

d. Find a unit vector in the direction of *w*.

Theorem 1: Let **u**, **v** and **w** be vectors in R^n, and let *c* be any scalar. Then

a. $u \bullet v = v \bullet u$ (symmetry axiom)
b. $(u+v) \bullet w = u \bullet w + v \bullet w$ (additivity axiom)
c. $(cu) \bullet v = c(u \bullet v) = u \bullet (cv)$ (homogeneity axiom)
d. $u \bullet u \geq 0$ and $u \bullet u = 0$ if and only if **u** = 0. (positivity axiom)

Length of a Vector

For $v = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_3 \end{bmatrix}$, the length or norm of **v** is the nonnegative scalar $\|v\|$ defined by

$\|v\| = \sqrt{v \bullet v} = \sqrt{v_1^2 + v_2^2 + \cdots + v_n^2}$ and $\|v\|^2 = v \bullet v$

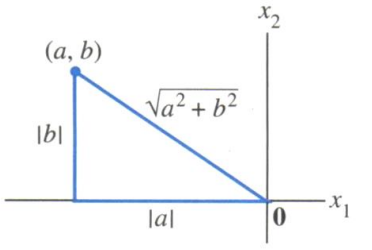


FIGURE 1
Interpretation of $\|\mathbf{v}\|$ as length.

For example, if $v = \begin{bmatrix} a \\ b \end{bmatrix}$, then $\|v\| = \sqrt{a^2 + b^2}$ (distance between **0** and **v**)

For any scalar c, $\|cv\| = |c| \|v\|$

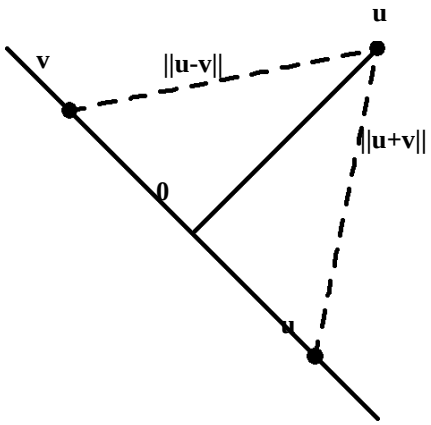
Distance in R^n The distance between u and v in R^n:- $dist(u,v) = \|u - v\|$

This agrees with the usual formulas for R^2 and R^3. Let $u = (u_1, u_2)$ and $v = (v_1, v_2)$.

Then **u** – **v** = $(u_1 - v_1, u_2 - v_2)$ and dist(**u**,**v**)= $\|u - v\| = \|(u_1 - v_1, u_2 - v_2)\| = \sqrt{(u_1 - v_1)^2 + (u_2 - v_2)^2}$

Example 2: Find the distance between $\begin{bmatrix} 1 \\ 3 \\ -3 \\ 4 \\ 0 \end{bmatrix}$ *and* $\begin{bmatrix} -2 \\ 1 \\ 0 \\ 4 \\ 3 \end{bmatrix}$

Orthogonal Vectors



From Geometry, we know that two lines are geometrically perpendicular iff the distance from **u** to **v** is the same as the distance from **u** to **–v**. This also means that the squares of the distances is the same.

$$\begin{aligned} [\text{dist}(\mathbf{u},\mathbf{v})]^2 &= \|\mathbf{u}-\mathbf{v}\|^2 = (\mathbf{u}-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) \\ &= (\mathbf{u})\cdot(\mathbf{u}-\mathbf{v})+(-\mathbf{v})\cdot(\mathbf{u}-\mathbf{v}) \\ &= \mathbf{u}\cdot\mathbf{u}-\mathbf{u}\cdot\mathbf{v}+-\mathbf{v}\cdot\mathbf{u}+\mathbf{v}\cdot\mathbf{v} \\ &= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2-2\mathbf{u}\cdot\mathbf{v} \\ \Rightarrow\ [\text{dist}(\mathbf{u},\mathbf{v})]^2 &= \|\mathbf{u}\|^2+\|\mathbf{v}\|^2-2\mathbf{u}\cdot\mathbf{v} \end{aligned}$$

Similarly, $[\text{dist}(\mathbf{u},-\mathbf{v})]^2 = \|\mathbf{u}\|^2+\|\mathbf{v}\|^2+2\mathbf{u}\cdot\mathbf{v}$

Since $[\text{dist}(\mathbf{u},-\mathbf{v})]^2 = [\text{dist}(\mathbf{u},\mathbf{v})]^2$, $\mathbf{u}\cdot\mathbf{v}$ = __?___.

Two vectors **u** and **v** are said to be orthogonal (to each other) if $\mathbf{u}\cdot\mathbf{v} = \mathbf{0}$.

Also note that if **u** and **v** are orthogonal, then $\|\mathbf{u}+\mathbf{v}\|^2 = \|\mathbf{u}\|^2+\|\mathbf{v}\|^2$.

Theorem 2 (The Pythagorean Theorem): Two vectors **u** and **v** are orthogonal if and only if $\|\mathbf{u}+\mathbf{v}\| = \|\mathbf{u}\|^2+\|\mathbf{v}\|^2$.

Example 3: Determine whether the given vectors are orthogonal.

a. $\mathbf{u} = (-1, 3, 2)$, $\mathbf{v} = (4, 2, -1)$

b. $\mathbf{u} = (u_1, u_1, u_1)$, $\mathbf{v} = (0, 0, 0)$

c. $\mathbf{u} = (0, 3, -2, 1)$, $\mathbf{v} = (5, 2, -1, 0)$

Orthogonal Complements

If a vector **z** is orthogonal to every vector in a subspace W of R^n, then **z** is said to be orthogonal to W. The set of vectors **z** that are orthogonal to W is called the orthogonal complement of W and is denoted by $W^{\perp}$ (read as "W perp").

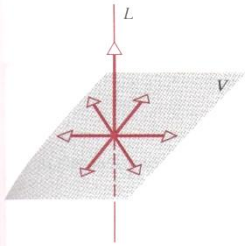


Figure 6.2.2
Every vector in L is orthogonal to every vector in V.

Row, Null and Columns Spaces

Theorem 3: Let A be an $m \times n$ matrix. Then the orthogonal complement of the row space of A is the nullspace of A, and the orthogonal complement of the column space of A is the nullspace of A^T

$$(\text{Row } A)^{\perp} = \text{Nul } A \text{ and}$$
$$(\text{Col } A)^{\perp} = \text{Nul } A^T$$

Ex 4: Let $A = \begin{bmatrix} 1 & 0 & -1 \\ -2 & 0 & 2 \end{bmatrix}$

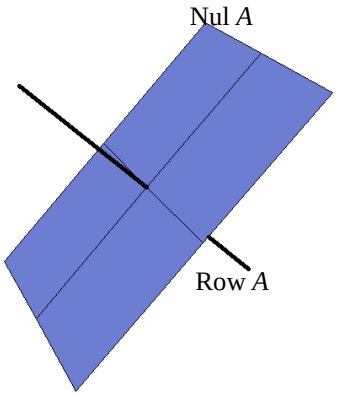
a. Find a basis for Nul A. Is Nul a line or plane in R^2 or R^3?

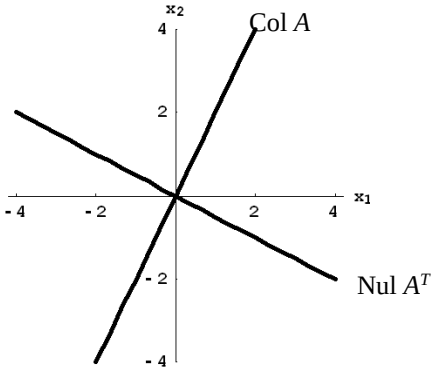
b. Find a basis for Row A. Is Row A a line or plane in R^2 or R^3?

c. Find a basis for Col A. Is Col A a line or plane in R^2 or R^3?

d. Find a basis for Nul A^T. Is Nul A^T a line or plane in R^2 or R^3?

Geometric representations of previous problem on Page 335.





6.2 Orthogonal Sets

A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ in R^n is called an orthogonal set if $\mathbf{u}_i \cdot \mathbf{u}_j = 0$ whenever $i \neq j$.

Ex 1: Is $\left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$ an orthogonal set?

Ex 2: Is $\left\{\begin{bmatrix}5\\-4\\0\\3\end{bmatrix},\begin{bmatrix}-4\\1\\-3\\8\end{bmatrix},\begin{bmatrix}3\\3\\5\\-1\end{bmatrix}\right\}$ an orthogonal set?

Theorem 4: Suppose $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthogonal set of nonzero vectors in R^n, then S is linearly independent and hence is a basis for the subspace spanned by S.
An orthogonal basis for a subspace W of R^n is a basis for W that is also an orthogonal set.

Ex 3: Show that $\mathbf{u}_1 = \begin{bmatrix}3\\-3\\0\end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix}2\\2\\-1\end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix}1\\1\\4\end{bmatrix}$ is an orthogonal basis for R^3.

Ex 4: Suppose $S = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthogonal basis for a subspace W of R^n and suppose $\mathbf{y}$ is in W. Find $c_1,\ldots,c_p$ so that $\mathbf{y} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + \cdots + c_p\mathbf{u}_p$.

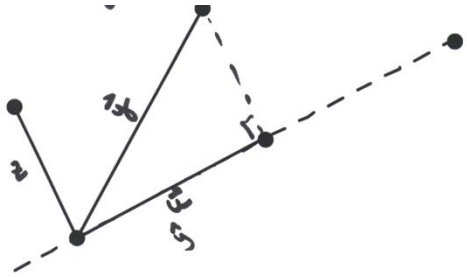
Hint: Begin with $\mathbf{y} \cdot \mathbf{u}_1 =$

Theorem 5: Let $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ be an orthogonal basis for a subspace W of R^n. Then each $\mathbf{y}$ in W has a unique representation as a linear combination of $\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p$. In fact, if

$\mathbf{y} = c_1\mathbf{u}_1 + \ldots + c_p\mathbf{u}_p$ then $c_j = \dfrac{\mathbf{y}\cdot\mathbf{u}_j}{\mathbf{u}_j\cdot\mathbf{u}_j} \quad (j = 1,\ldots, p)$

Ex 5: Express $\mathbf{y} = \begin{bmatrix}3\\7\\4\end{bmatrix}$ as a linear combination of the orthogonal basis $\left\{\begin{bmatrix}1\\-1\\0\end{bmatrix},\begin{bmatrix}1\\1\\0\end{bmatrix},\begin{bmatrix}0\\0\\1\end{bmatrix}\right\}$. Hint: Find c_1, c_2, and c

Orthogonal Projections



For a nonzero vector **u** in R^n, suppose we want to write **y** in R^n as the following

y= (multiple of **u**) + (multiple of a vector $\perp$ to **u**)
y= α**u** + **z** $\Rightarrow$ **z** = **y** - α**u.**

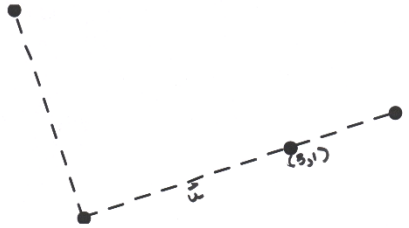
So, $\mathbf{z}\cdot\mathbf{u} = 0 \Rightarrow (\mathbf{y} - \alpha\mathbf{u})\cdot\mathbf{u} = 0 \Rightarrow \mathbf{y}\cdot\mathbf{u} - \alpha(\mathbf{u}\cdot\mathbf{u}) = 0 \Rightarrow \alpha = \dfrac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}$

$\hat{\mathbf{y}} = \dfrac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}$ is called the orthogonal projection of **y** onto **u** & $\mathbf{z} = \mathbf{y} - \dfrac{\mathbf{y}\cdot\mathbf{u}}{\mathbf{u}\cdot\mathbf{u}}\mathbf{u}$ *or* $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$ is called the component of **y** orthogonal to **u**

Ex 6: Compute the orthogonal projection of $\begin{bmatrix} 1 \\ -1 \end{bmatrix}$ onto the line through $\begin{bmatrix} -1 \\ 3 \end{bmatrix}$ and the origin.

Ex 7: Let $\mathbf{y} = \begin{bmatrix} 2 \\ 6 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 7 \\ 1 \end{bmatrix}$. Write **y** as the sum of a vector in the Span{**u**} and a vector orthogonal to **u**.

Ex 8: Let $\mathbf{y} = \begin{bmatrix} -8 \\ 4 \end{bmatrix}$ and $\mathbf{u} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Compute the distance from **y** to the line through **0** and **u**.



Orthonormal Sets
A set of vectors $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ in R^n is called an orthonormal set if it is an orthogonal set of unit vectors.

If $W = \text{span}\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$, then $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthonormal basis for *W*.

Ex 9: Determine which sets are orthonormal. If a set is only orthogonal, normalize the vectors to produce an orthonormal set.

a. $\begin{bmatrix} -0.6 \\ 0.8 \end{bmatrix}, \begin{bmatrix} 0.8 \\ 0.6 \end{bmatrix}$

b. $\begin{bmatrix} -\frac{2}{3} \\ \frac{1}{3} \\ \frac{2}{3} \end{bmatrix}, \begin{bmatrix} \frac{1}{3} \\ \frac{2}{3} \\ 0 \end{bmatrix}$

Recall that **v** is a unit vector if $\|v\| = \sqrt{v \cdot v} = \sqrt{v^T v} = 1$.

Suppose that $U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \mathbf{u}_3]$ where $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal set.

Then, $U^T U = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \mathbf{u}_3^T \end{bmatrix} \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \mathbf{u}_3 \end{bmatrix} =$

$\begin{bmatrix} \mathbf{u}_1^T \mathbf{u}_1 & _____ & _____ \\ _____ & _____ & _____ \\ _____ & _____ & _____ \end{bmatrix}$ $\mathbf{u}_1^T \ \mathbf{u}_1 = 1$ and _____ = 1 and _____ = 1

But all others are _____ so the dot product is _____ resulting in the matrix

Therefore, we have shown that $U^T U = I$. So, $U^{-1} = U^T$(such a matrix is called an orthogonal matrix.)

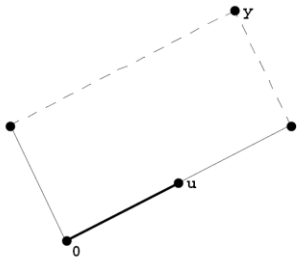
Theorem 6: An *m* x *n* matrix *U* has orthonormal columns if and only if $U^T U = I$.

Theorem 7: Let *U* be an *m* x *n* matrix with orthonormal columns, and let **x** and **y** be in R^n. Then

a. $\|U\mathbf{x}\| = \|\mathbf{x}\|$ b. $(U\mathbf{x}) \cdot (U\mathbf{y}) = \mathbf{x} \cdot \mathbf{y}$ c. $(U\mathbf{x}) \cdot (U\mathbf{y}) = 0$ if and only if $\mathbf{x} \cdot \mathbf{y} = 0$.

6.3 Orthogonal Sets

Review: $\hat{\mathbf{y}} = \dfrac{\mathbf{y} \cdot \mathbf{u}}{\mathbf{u} \cdot \mathbf{u}} \mathbf{u}$ is the orthogonal projection of _____ onto _____.



Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthogonal basis for *W* in R^n. For each **y** in *W*,

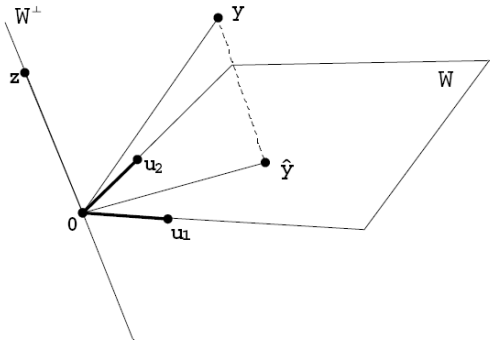
$$\mathbf{y} = \left(\frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \right) u_1 + \cdots + \left(\frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p} \right) \mathbf{u}_p$$

Ex 1: Suppose $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthogonal basis for R^3 and let $W = \text{Span}\{\mathbf{u}_1, \mathbf{u}_2\}$. Write $\mathbf{y}$ in R^3 as the sum of a vector $\hat{\mathbf{y}}$ in W and a vector $\mathbf{z}$ in $W^{\perp}$.

To show that $\mathbf{z}$ is orthogonal to every vector in W, show that $\mathbf{z}$ is orthogonal to the vectors in$\{\mathbf{u}_1, \mathbf{u}_2\}$.

Since $\mathbf{z} \cdot \mathbf{u}_1 =$ $\mathbf{z} \cdot \mathbf{u}_2 =$

Ex 2: If $\mathbf{u}_1 = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} -2 \\ 1 \\ -1 \\ 1 \end{bmatrix}, \mathbf{u}_3 = \begin{bmatrix} 1 \\ 1 \\ -2 \\ -1 \end{bmatrix}, \mathbf{u}_4 = \begin{bmatrix} -1 \\ 1 \\ 1 \\ -2 \end{bmatrix}, \mathbf{v} = \begin{bmatrix} 4 \\ 5 \\ -3 \\ 3 \end{bmatrix}$ write $\mathbf{v}$ as the sum of two vectors, one in Span$\{\mathbf{u}_1\}$ and the other in Span$\{\mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$.

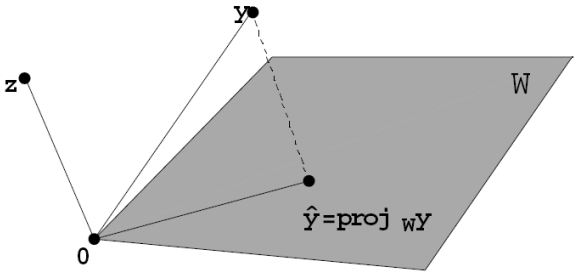


Theorem 8 – The Orthogonal Decomposition Theorem: Let W be a subspace of R^n. Then each $\mathbf{y}$ in R^n can be uniquely represented in the form $\mathbf{y} = \hat{\mathbf{y}} + \mathbf{z}$ where $\hat{\mathbf{y}}$ is in W and $\mathbf{z}$ is in $W^{\perp}$. In fact, if $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is any orthogonal basis of W, then $\hat{\mathbf{y}} = \left(\frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1}\right) u_1 + \cdots + \left(\frac{\mathbf{y} \cdot \mathbf{u}_p}{\mathbf{u}_p \cdot \mathbf{u}_p}\right) \mathbf{u}_p$ and $\mathbf{z} = \mathbf{y} - \hat{\mathbf{y}}$.

The vector $\hat{\mathbf{y}}$ is called the orthogonal projection of y onto W.

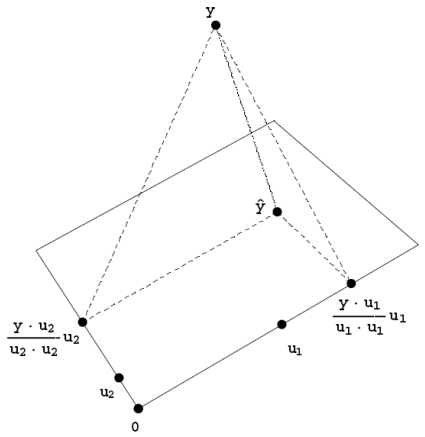
Ex 3: Verify that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal set, and then find the orthogonal projection of $\mathbf{y}$ onto Span$\{\mathbf{u}_1, \mathbf{u}_2\}$ given that

$\mathbf{u}_1 = \begin{bmatrix} -4 \\ -1 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$, and $\mathbf{y} = \begin{bmatrix} 6 \\ 4 \\ 1 \end{bmatrix}$.



Ex 4: Let $\mathbf{u}_1 = \begin{bmatrix} 3 \\ 0 \\ 1 \end{bmatrix}, \mathbf{u}_2 = \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}$, and $\mathbf{y} = \begin{bmatrix} 0 \\ 3 \\ 10 \end{bmatrix}$. Observe that $\{\mathbf{u}_1, \mathbf{u}_2\}$ is an orthogonal basis for W = Span$\{\mathbf{u}_1, \mathbf{u}_2\}$.

Write $\mathbf{y}$ as the sum of a vector in W and a vector orthogonal to W. Hint: Begin with $\text{proj}_W \mathbf{y} = \hat{\mathbf{y}} = \left(\frac{\mathbf{y} \cdot \mathbf{u}_1}{\mathbf{u}_1 \cdot \mathbf{u}_1} \right) \mathbf{u}_1 + \left(\frac{\mathbf{y} \cdot \mathbf{u}_2}{\mathbf{u}_2 \cdot \mathbf{u}_2} \right) \mathbf{u}_2$



Geometric Interpretation of Orthogonal Projections

Theorem 9 – The Best Approximation Theorem

Let W be a subspace of R^n, $\mathbf{y}$ any vector in R^n, and $\hat{\mathbf{y}}$ the orthogonal projection of $\mathbf{y}$ onto W. Then $\hat{\mathbf{y}}$ is the point in W closest to $\mathbf{y}$, in the sense that

$$\|\mathbf{y} - \hat{\mathbf{y}}\| < \|\mathbf{y} - \mathbf{v}\|$$

for all $\mathbf{v}$ in W distinct from $\hat{\mathbf{y}}$.

Ex 5: Find the closest point to $\mathbf{y}$ in Span$\{\mathbf{u}_1, \mathbf{u}_2\}$. where $\mathbf{y} = \begin{bmatrix} 2 \\ 4 \\ 0 \\ -2 \end{bmatrix}, \mathbf{u}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}$, and $\mathbf{u}_2 = \begin{bmatrix} 0 \\ 0 \\ 1 \\ 1 \end{bmatrix}$

Part of Theorem 10 below is based upon another way to view matrix multiplication where A is m x p and B is p x n.

$$AB = \begin{bmatrix} col_1 A & col_2 A & \cdots & col_p A \end{bmatrix} \begin{bmatrix} row_1 B \\ row_2 B \\ \vdots \\ row_p B \end{bmatrix} = (col_1 A)(row_1 B) + \cdots + (col_p A)(row_p B)$$

For example,

$$\begin{bmatrix} 5 & 6 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 34 & 5 & 3 \\ 10 & 3 & 7 \end{bmatrix}$$ Is the same as: $$\begin{bmatrix} 5 & 6 \\ 3 & 1 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \\ 4 & 0 & -2 \end{bmatrix} = \begin{bmatrix} 5 \\ 3 \end{bmatrix} \begin{bmatrix} 2 & 1 & 3 \end{bmatrix} + \begin{bmatrix} 6 \\ 1 \end{bmatrix} \begin{bmatrix} 4 & 0 & -2 \end{bmatrix}$$

So if $U = \begin{bmatrix} \mathbf{u}_1 & \mathbf{u}_2 & \cdots & \mathbf{u}_p \end{bmatrix}$. Then $U^T = \begin{bmatrix} \mathbf{u}_1^T \\ \mathbf{u}_2^T \\ \vdots \\ \mathbf{u}_p^T \end{bmatrix}$.

So, …

Theorem 10: If $\{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_p\}$ is an orthornormal basis for a subspace W of R^n, then

$$\text{proj}_W \mathbf{y} = (\mathbf{y} \cdot \mathbf{u}_1)\mathbf{u}_1 + \cdots + (\mathbf{y} \cdot \mathbf{u}_p)\mathbf{u}_p$$

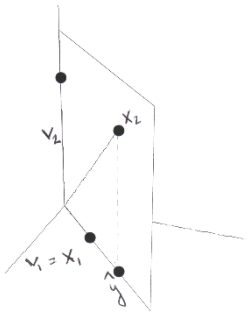
If $U = [\mathbf{u}_1 \quad \mathbf{u}_2 \quad \ldots \mathbf{u}_p]$, then $\text{proj}_W \mathbf{y} = UU^T\mathbf{y}$ for all $\mathbf{y}$ in R^n.

6.4 The Gram-Schmidt Process

Goal:Form an orthogonal basis for a subspace W.

Ex 1: Suppose $W = \text{Span}\{\mathbf{x}_1, \mathbf{x}_2\}$ where $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ and $\mathbf{x}_2 = \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}$. Find an orthogonal basis $\{\mathbf{v}_1, \mathbf{v}_2\}$. for W.

Drawing:



Begin with $\mathbf{v}_1 = \mathbf{x}_1 = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$

Reminder: $\hat{\mathbf{y}} = \text{proj}_{\mathbf{v}_1} \mathbf{x}_2 = \left(\dfrac{x_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1$

And, $\mathbf{v_2} = \mathbf{x}_2 - \hat{\mathbf{y}} = \mathbf{x}_2 - \left(\dfrac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \right) \mathbf{v}_1 = \ldots..$

Ex 2: Suppose $\{\mathbf{x}_1, \mathbf{x}_2, \mathbf{x}_3\}$ is a basis for a subspace W of R^4. Describe an orthogonal basis for W.

Note that $\mathbf{v}_3$ is in W. Why?

$\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is an orthogonal basis for W.

Theorem 11 – The Gram-Schmidt Process: Given a basis $\{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_p\}$ for a subspace W of R^n, define

$$\mathbf{v}_1 = \mathbf{x}_1$$

$$\mathbf{v_2} = \mathbf{x}_2 - \frac{\mathbf{x}_2 \cdot \mathbf{v}_1}{\mathbf{v}_1 \cdot \mathbf{v}_1} \mathbf{v}_1$$

$$\mathbf{v_3} = \mathbf{x_3} - \frac{\mathbf{x_3} \cdot \mathbf{v_1}}{\mathbf{v_1} \cdot \mathbf{v_1}} \mathbf{v_1} - \frac{\mathbf{x_3} \cdot \mathbf{v_2}}{\mathbf{v_2} \cdot \mathbf{v_2}} \mathbf{v}_2$$

$$\vdots$$

$$\mathbf{v_p} = \mathbf{x_p} - \frac{\mathbf{x_p} \cdot \mathbf{v_1}}{\mathbf{v_1} \cdot \mathbf{v_1}} \mathbf{v_1} - \frac{\mathbf{x_p} \cdot \mathbf{v_2}}{\mathbf{v_2} \cdot \mathbf{v_2}} \mathbf{v_2} - \ldots - \frac{\mathbf{x}_p \mathbf{v}_{p\text{-}1}}{\mathbf{v}_{p\text{-}1} \cdot \mathbf{v}_{p\text{-}1}} \mathbf{v}_{p\text{-}1}$$

Then $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$ is an orthogonal basis for W and Span $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ = Span $\{\mathbf{v}_1, \ldots, \mathbf{v}_p\}$

Ex 3:

Suppose $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$ where $\mathbf{x}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \\ 0 \end{bmatrix}, \mathbf{x}_2 = \begin{bmatrix} 1 \\ 2 \\ 0 \\ 0 \end{bmatrix}, \mathbf{x}_3 = \begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}$, is a basis for a subspace W of R^4. Describe an orthogonal basis for W.

Orthonormal Basis

Suppose the following is an orthogonal basis for subspace $W = \text{span}\left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 2 \\ 2 \\ 3 \end{bmatrix}\right\}$: GS used to produce… $\{\mathbf{v}_1, \mathbf{v}_2\} = \left\{\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix}\right\}$

Rescale to form unit vectors: $\mathbf{u}_1 = \frac{1}{\|\mathbf{v}_1\|}\mathbf{v}_1 = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} \\ 1/\sqrt{2} \\ 0 \end{bmatrix}$ $\mathbf{u}_2 = \frac{1}{\|\mathbf{v}_2\|}\mathbf{v}_2 = \frac{1}{3}\begin{bmatrix} 0 \\ 0 \\ 3 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}$

Ex 4: Use the GS process then normalize to transform the basis $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3, \mathbf{u}_4\}$ into an orthonormal basis. $\mathbf{u}_1 = (0, 2, 1, 0)$, $\mathbf{u}_2 = (1, -1, 0, 0)$, $\mathbf{u}_3 = (1, 2, 0, -1)$ and $\mathbf{u}_4 = (1, 0, 0, 1)$

QR Factorization

If A is an m x n matrix with linearly independent column vectors, and if Q is the matrix with orthonormal column vectors that results from applying the Gram-Schmidt process to the column vectors of A, what relationship exists between A and Q?

The answer lies in the following theorem.

Theorem 12 (The QR Factorization Theorem)

If A is an m x n matrix with linearly independent columns, then A can be factored as $A = QR$, where Q is an m x n matrix whose columns form an orthogonal basis for Col A and R is an n x n upper triangular invertible matrix with positive entries on its main diagonal.

So, finding Q is simple—it is the column matrix of our orthonormal vectors. How do we find R?

Remember: Since Q has orthonormal columns $Q^TQ = I$. So, if

$A = QR$

$Q^TA = Q^TQR$

$Q^TA = R$

Ex 5: Find the QR factorization of $A = \begin{bmatrix} 1 & 2 \\ 1 & 2 \\ 0 & 3 \end{bmatrix}$

Ex 6: Find the QR factorization of $A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix}$.

What is the role of the QR-decomposition in Linear Algebra. In recent years the QR-decomposition has assumed growing importance as the mathematical foundation for a wide variety of practical numerical algorithms, including a widely used algorithm for computing eigenvalues of large matrices. Such algorithms are discussed in books that deal with advanced numerical methods of linear algebra.

6.5 Least-Squares Problem

Problem: What do we do when $A\mathbf{x} = \mathbf{b}$ has no solution $\mathbf{x}$?

Answer: Find $\hat{\mathbf{x}}$ such that $A\hat{\mathbf{x}}$ is as "close" as possible to $\mathbf{b}$. (Least Squares Problem)

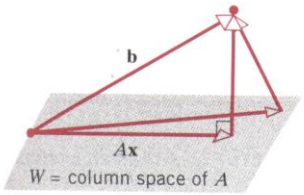
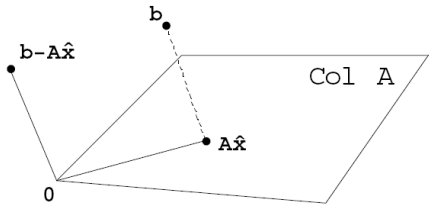


Figure 6.4.2
A least squares solution $\mathbf{x}$ produces the vector $A\mathbf{x}$ in W closest to $\mathbf{b}$.

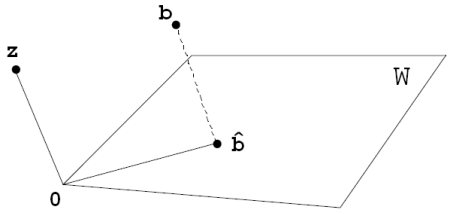
If A is m x n and $\mathbf{b}$ is in R^m, a least squares solution of $A\mathbf{x} = \mathbf{b}$ is an $\hat{\mathbf{x}}$ in R^m such that $\|\mathbf{b} - A\hat{\mathbf{x}}\| \le \|\mathbf{b} - A\mathbf{x}\|$ for all $\mathbf{x}$ in R^n.



Let W = Col A where A is m x n and $A = [\mathbf{a}_1 \ \mathbf{a}_2 \ \ldots \ \mathbf{a}_n]$. Suppose $\mathbf{b}$ is in R^m and $\hat{\mathbf{b}} = \text{proj}_W \mathbf{b}$.

$\hat{\mathbf{b}}$ is the point in W = Col A closest to $\mathbf{b}$

Since $\hat{\mathbf{b}}$ is in Col A, then $\hat{\mathbf{x}}$ is a vector in R^n such that $\hat{\mathbf{b}} = A\hat{\mathbf{x}}$



By the Orthogonal Projection Theorem, $\mathbf{z}$ is in $W^{\perp}$ where $\mathbf{z} = \mathbf{b} - A\hat{\mathbf{x}}$.

Then $\mathbf{b} - A\hat{\mathbf{x}}$ is orthogonal to every column of A. Meaning that

$$\mathbf{a}_1^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0$$
$$\mathbf{a}_2^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0 \quad \ldots.$$
$$\mathbf{a}_n^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0$$
$$\begin{bmatrix} \mathbf{a}_1^T \\ \mathbf{a}_2^T \\ \vdots \\ \mathbf{a}_n^T \end{bmatrix}(\mathbf{b} - A\hat{\mathbf{x}}) = \begin{bmatrix} 0 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$A^T(\mathbf{b} - A\hat{\mathbf{x}}) = 0$$
$$A^T\mathbf{b} - A^TA\hat{\mathbf{x}} = 0$$

$$\boxed{A^TA\hat{\mathbf{x}} = A^T\mathbf{b}}$$

(normal equations for $\hat{\mathbf{x}}$)

Theorem 13: The set of least squares solutions of $A\mathbf{x} = \mathbf{b}$ is the set of all solutions of the normal equations $A^TA\hat{\mathbf{x}} = A^T\mathbf{b}$.

Ex 1: Find a least squares solution to the inconsistent system $A\mathbf{x} = \mathbf{b}$ where $A = \begin{bmatrix} 2 & 0 \\ 0 & 1 \\ 2 & 2 \end{bmatrix}$ and $\mathbf{b} = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$.

When $A^T A$ is invertible,

$$A^T A\hat{\mathbf{x}} = A^T \mathbf{b}$$
$$(A^T A)^{-1} A^T A\hat{\mathbf{x}} = \left(A^T A\right)^{-1} A^T \mathbf{b}$$
$$\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$$

Ex 2: Do Example 1 using the inverse instead.

Ex 3: Describe all least-squares solutions of the equation $A\mathbf{x} = \mathbf{b}$ $A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \\ 1 & 0 & 1 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 7 \\ 2 \\ 3 \\ 6 \\ 5 \\ 4 \end{bmatrix}$

Theorem 14: The matrix $A^T A$ is invertible if and only if the columns of A are linearly independent. In this case, the equation $A\mathbf{x} = \mathbf{b}$ has only one least-squares solution $\hat{\mathbf{x}}$, and it is given by $\hat{\mathbf{x}} = (A^T A)^{-1} A^T \mathbf{b}$.

There is an error in doing least-squares and it is $\|\mathbf{b} - A\hat{\mathbf{x}}\|$ (least-squares error)

Ex 4: Calculate the least-squares error from Ex 2.

Alternative Calculations of Least-Squares Solutions

The next example shows how to find a least-squares solution of $A\mathbf{x} = \mathbf{b}$ when the columns of A are orthogonal.

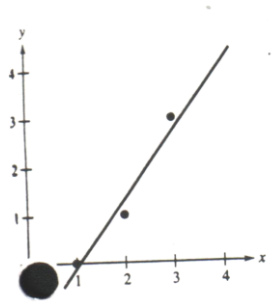
Ex 5: Find (a) the orthogonal projection of $\mathbf{b}$ onto Col A and (b) a least-squares solution of $A\mathbf{x} = \mathbf{b}$. $A = \begin{bmatrix} 1 & 2 \\ -1 & 4 \\ 1 & 2 \end{bmatrix}, \mathbf{b} = \begin{bmatrix} 3 \\ -1 \\ 5 \end{bmatrix}$

Ex 6: Find the least squares solution of the linear system $A\mathbf{x} = \mathbf{b}$ given by

$$\begin{aligned} x_1 - x_2 &= 4 \\ 3x_1 + 2x_2 &= 1 \\ -2x_1 + 4x_2 &= 3 \end{aligned}$$

6.6 Least-Squares Lines

Experimental data often produce $(x_1, y_1), \ldots, (x_n, y_n)$ that, when graphed, seem to lie close to a line.



Consider the following example. Suppose you are given three points (1, 0), (2,1) and (3,3). How can you find the line $y = \beta_0 + \beta_1 x$ that best fits these points?

We know that if the points were collinear that the system

$$\begin{aligned} \beta_0 + \beta_1 &= 0 \\ \beta_0 + 2\beta_1 &= 1 \\ \beta_0 + 3\beta_1 &= 3 \end{aligned}$$

would be consistent.

We can write the system as $X\beta = \mathbf{y}$ where

$$X = \begin{bmatrix} 1 & 1 \\ 1 & 2 \\ 1 & 3 \end{bmatrix}, \ \mathbf{y} = \begin{bmatrix} 0 \\ 1 \\ 3 \end{bmatrix} \text{ and } \beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$$

The system is inconsistent. We cannot find a β such that $X\beta = \mathbf{y}$ but we can look for a β that minimizes the norm of the error $\| X\beta - \mathbf{y} \|$ using the techniques from the previous section.

The solution $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$ to this minimization problem is called the least squares line and has the form $y = \beta_0 + \beta_1 x$. This line is also called a line of regression of y on x, because any errors in the data are assumed to be only in the y-coordinates. The coefficients β_0 and β_1 are called the regression coefficients.

Generalizing, we want to determine the parameters β_0 and β_1 that make the line as "close" to the points as possible.

Corresponding to each data point (x_j, y_j) there is a point $(x_j, \beta_o + \beta_1 x_j)$ on the line with the same x-coordinate (see diagram on p 419)

We call y_j the observed value of y and $\beta_0 + \beta_1 x_j$ the predicted y-value (determined by the line). The difference between an observed y-value and a predicted y–value is called a residual.
There are several ways to measure how "close" the line is to the data. The usual choice (primarily because the mathematical calculations are simple) is to add the squares of the residuals.

The least-squares line is the line $y = \beta_0 + \beta_1 x$ that minimizes the sum of the squares of the residuals.

If the data points were points were on the line, the parameters β_0 and β_1 would satisfy the equations

Predicted y-value		Observed y-value
$\beta_0 + \beta_1 x_1$	=	y_1
$\beta_0 + \beta_1 x_2$	=	y_2
$\vdots$		$\vdots$
$\beta_0 + \beta_1 x_n$	=	y_n

We can write the system as $x\beta = y$, where $X = \begin{bmatrix} 1 & x_1 \\ 1 & x_2 \\ \vdots & \vdots \\ 1 & x_n \end{bmatrix}$, $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$

Of course, if the data points do not lie on a line, then there are no parameters β_0, β_1 for which the predicted y-values in $X\beta$ equal the observed *y*-values in **y**, and $X\beta = \mathbf{y}$ has no solution.
This is a least-square problem, *A***x** = **b**, with different notation!

The square of the distance between the vectors $X\beta$ and **y** is precisely the sum of the squares of the residuals.

The β that minimizes this sum also minimizes the distance between $X\beta$ and **y.**

Computing the least-squares solution of $X\beta = \mathbf{y}$ is equivalent to finding the β that determines the least-squares line in the previous picture.

For the least-squares solution of $X\beta = \mathbf{y}$, obtain the normal equations (with the new notation): $X^T X\beta = X^T \mathbf{y}$

Ex 1: Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the given data points. (1, 0), (2, 1), (4, 2), (5, 3)

Ex 2: Find the equation $y = \beta_0 + \beta_1 x$ of the least-squares line that best fits the given data points.(2, 3), (3, 2), (5, 1), (6, 0)

This process can be extended to find a "curve" of best fit. Suppose the points appear to lie along a parabola rather than a line. We wish to approximate the data by an equation of the form $y = \beta_0 + \beta_1 x + \beta_2 x^2$. We follow the same process but

$X = \begin{bmatrix} 1 & x_1 & x_1^2 \\ 1 & x_2 & x_2^2 \\ \vdots & \vdots & \vdots \\ 1 & x_n & x_n^2 \end{bmatrix}$, $\beta = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \beta_2 \end{bmatrix}$, $\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix}$

Note: Hold #3-4 for after the worksheet.

Ex 3: Find the least squares quadratic polynomial fit to the four (-1, 1), (0, 0), (1,1) and (2, 2).

Least squares problems plan a fundamental role in mathematical modeling of real life phenomena.
Ex 4: The following table gives the world population in billions for seven different years

Year	1975	1980	1985	1990	1995	2000	2005
Population	4.1	4.5	4.9	5.3	5.7	6.1	6.5

Find the least squares regression linear model for these data and use the model to estimate the population for the year 2020.

6.7 Inner Product Spaces

Notions of length, distance, and orthogonality are often important in applications involving a vector space.

We are familiar with the dot product and its properties:

$\mathbf{u} \cdot \mathbf{v} = \mathbf{v} \cdot \mathbf{u}$
$(\mathbf{u} + \mathbf{v}) \cdot \mathbf{w} = \mathbf{u} \cdot \mathbf{w} + \mathbf{v} \cdot \mathbf{w}$
$(k\mathbf{u}) \cdot \mathbf{v} = k(\mathbf{u} \cdot \mathbf{v})$
$\mathbf{v} \cdot \mathbf{v} \geq 0$ where $\mathbf{v} \cdot \mathbf{v} = 0$ iff $\mathbf{v} = \mathbf{0}$

We will extend this idea—give it a new name (inner product) and a new symbol $\langle \mathbf{u}, \mathbf{v} \rangle$.

Definition: An inner product on a vector space V is a function that, to each pair of vectors $\mathbf{u}$ and $\mathbf{v}$ in V, associates a real number $\langle \mathbf{u}, \mathbf{v} \rangle$ and satisfies the following axioms, for all $\mathbf{u}$, $\mathbf{v}$, $\mathbf{w}$ in V and all scalars c:

1. Symmetry: $\langle \mathbf{u}, \mathbf{v} \rangle = \langle \mathbf{v}, \mathbf{u} \rangle$
2. Additivity: $\langle \mathbf{u} + \mathbf{v}, \mathbf{w} \rangle = \langle \mathbf{u}, \mathbf{w} \rangle + \langle \mathbf{v}, \mathbf{w} \rangle$
3. Homogeneity: $\langle c\mathbf{u}, \mathbf{v} \rangle = c \langle \mathbf{u}, \mathbf{v} \rangle$
4. Positivity: $\langle \mathbf{u}, \mathbf{u} \rangle \geq 0$ and $\langle \mathbf{u}, \mathbf{u} \rangle = 0$ if and only if $\mathbf{u} = 0$

A real vector space with an inner product is called an inner product space.

Ex 1: Let $\mathbf{u} = (\mathbf{u}_1, \mathbf{u}_2)$ and $\mathbf{v} = (\mathbf{v}_1, \mathbf{v}_2)$ in R^2. Show the function $\langle \mathbf{u}, \mathbf{v} \rangle = 2\mathbf{u}_1\mathbf{v}_1 + 5\mathbf{u}_2\mathbf{v}_2$ is an inner product on R^2. (Hint: You must show that the four axioms are true.)

Ex 2: Let $\mathbf{u} = (\mathbf{u}_1, \mathbf{u}_2)$ and $\mathbf{v} = (\mathbf{v}_1, \mathbf{v}_2)$ in R^2. Show the function $\langle \mathbf{u}, \mathbf{v} \rangle = 2\mathbf{u}_1\mathbf{v}_1 - 5\mathbf{u}_2\mathbf{v}_2$ is NOT an inner product on R^2.

The vector space R^n with the standard inner product (dot product) is an inner product space, and almost everything discussed for the dot product over R^n carries over to inner product spaces.

Lengths, Distances, and Orthogonality

Let V be an inner product space, with the inner product denoted by $\langle \mathbf{u}, \mathbf{v} \rangle$. Just as in R^n, we define the length, or norm, of a vector $\mathbf{v}$ to be the scalar $\|\mathbf{v}\| = \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}$.

Equivalently, $\|\mathbf{v}\|^2 = \langle \mathbf{v}, \mathbf{v} \rangle$, however this definition does not say that $\langle \mathbf{v}, \mathbf{v} \rangle$ is a "sum of squares" because v need not be an element of R^n. A unit vector is one whose length is 1.

The distance between $\mathbf{u}$ and $\mathbf{v}$ is $\|\mathbf{u} - \mathbf{v}\| = \sqrt{(\mathbf{u} - \mathbf{v}) \bullet (\mathbf{u} - \mathbf{v})}$.

Vectors $\mathbf{u}$ and $\mathbf{v}$ are orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

Special Inner Products

Weighted Inner Products—weights are given to the various entries in the sum for the inner product in such a way that more importance is given to the more reliable measurements

Ex 3: Let $\mathbf{u} = (1, -3)$ and $\mathbf{v} = (5, 1)$. Use the linear inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 3\mathbf{u}_1\mathbf{v}_1 + 5\mathbf{u}_2\mathbf{v}_2$ to compute

a. $\langle \mathbf{u}, \mathbf{v} \rangle$

b. $\|\mathbf{u}\|$

c. $\|\mathbf{v}\|$

d. $\|\mathbf{u} - \mathbf{v}\|$

Polynomial Inner Products—for p and q, the inner product is defined to be $\langle p,q\rangle = p(t_0)q(t_0) + p(t_1)q(t_1) + \ldots + p(t_n)q(t_n)$

Ex 4: Given $p(t) = 3t - t^2$, $q(t) = 3 + 2t^2$ and $\langle p,q\rangle$ is defined as evaluation of the polynomials at -1, 0, and 1. Find

a. $\langle p,q\rangle$

b. $\|p\|$

c. $\|q\|$

d. $\|p - q\|$

Inner Products Generated by Matrices—Given $\mathbf{u} = \begin{bmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{bmatrix}$ and A is an invertible n x n matrix then $\langle \mathbf{u}, \mathbf{v}\rangle = A\mathbf{u} \cdot A\mathbf{v}$

Ex 5: Let $\mathbf{u} = \begin{bmatrix} 5 \\ 2 \end{bmatrix}$ and $\mathbf{v} = \begin{bmatrix} 3 \\ 1 \end{bmatrix}$. Use the inner product on R^2 generated by the matrix $\begin{bmatrix} 1 & -1 \\ 0 & 2 \end{bmatrix}$ to find

a. $\langle \mathbf{u}, \mathbf{v}\rangle$

b. $\|\mathbf{u}\|$

c. $\|\mathbf{v}\|$

d. $\|\mathbf{u} - \mathbf{v}\|$

Calculus Inner Product—$C[a, b]$. Given two continuous functions f and g over the interval $[a, b]$, $\langle f, g\rangle = \int_a^b f(x)g(x)dx$

Ex 6: Given $f(t) = 5t - 3$, $g(t) = t^3 - t^2$ on $C[0, 1]$, find

a. $\langle f, g\rangle$

b. $\|g\|$

Ex 7: Given $f(t) = 3t - 5, \quad g(t) = t^2$ on $C[0, 1]$, find

a. $\langle f, g \rangle$

b. $\|f\|$

c. $\|g\|$

The Gram-Schmidt Process

The existence of orthogonal bases for finite-dimensional subspaces of an inner product space can be established by the Gram-Schmidt process, just as in R^n. The orthogonal projection of a vector onto a subspace W with an orthogonal basis can be constructed as usual.

The projection does not depend on the choice of orthogonal basis, and it has the properties described in the Orthogonal Decomposition Theorem and the Best Approximation Theorem.

Ex 8: Compute the orthogonal projection of q onto the subspace spanned by p, for p and q in Ex 4: $p(t) = 3t - t^2, \quad q(t) = 3 + 2t^2$

Two Inequalities

Given a vector $\mathbf{v}$ in an inner product space V and given a finite-dimensional subspace W, we may apply the Pythagorean Theorem to the orthogonal decomposition of $\mathbf{v}$ with respect to W and obtain $\|\mathbf{v}\|^2 = \|proj_W \mathbf{v}\|^2 + \|\mathbf{v} - proj_W \mathbf{v}\|^2$.
The norm of the projection of $\mathbf{v}$ onto W does not exceed the norm of $\mathbf{v}$ itself. This simple observation leads to the following important inequality.

Theorem 16: The Cauchy-Schwarz Inequality: For all $\mathbf{u}, \mathbf{v}$ in V, $|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\| \|\mathbf{v}\|$

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \le \|\mathbf{u}\| \|\mathbf{v}\| \Rightarrow$$

If we start with this inequality, $\dfrac{|\langle \mathbf{u}, \mathbf{v} \rangle|}{\|\mathbf{u}\| \|\mathbf{v}\|} \le 1 \Rightarrow$

$$-1 \le \frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\|\mathbf{u}\| \|\mathbf{v}\|} \le 1$$

The cosine of the angle between the two vectors is $\cos\theta = \dfrac{\langle \mathbf{u}, \mathbf{v} \rangle}{\| \mathbf{u} \| \| \mathbf{v} \|}$. So, the Cauchy-Schwarz inequality guarantees that the $\cos\theta$ is between -1 and 1 no matter what dimension R^n. It has true geometric meaning in R^2 and R^3. Our primary interest will be to determine the orthogonality of given vectors.

When we define $\cos\theta = \dfrac{\langle \mathbf{u}, \mathbf{v} \rangle}{\| \mathbf{u} \| \| \mathbf{v} \|}$, our θ is defined to be on the interval [0, π].

The Cauchy-Schwarz inequality is useful in many branches of mathematics. Our main need for it here is to prove another fundamental inequality involving norms of vectors.

Ex 9: Let **u** = (7, -1), **v** = (2, 3) and $\langle \mathbf{u}, \mathbf{v} \rangle = 3\mathbf{u}_1\mathbf{v}_1 + 5\mathbf{u}_2\mathbf{v}_2$. Find the $\cos\theta$ between **u** and **v**.

Ex 10: Let $p(t) = 3t - t^2$, $q(t) = 3 + 2t^2$ and $\langle p, q \rangle$ is defined as evaluation of the polynomials at -1, 0, and 1. Find the $\cos\theta$ between *p* and *q*.

Two vectors **u** and **v** are orthogonal if $\langle \mathbf{u}, \mathbf{v} \rangle = 0$

Theorem 17 The Triangle Inequality

For all **u**, **v** in *V*, $\|\mathbf{u} + \mathbf{v}\| \le \|\mathbf{u}\| + \|\mathbf{v}\|$

We can also generalize the Pythagorean Theorem and say

If **u** and **v** are orthogonal vectors in an inner product space then $\|\mathbf{u}\|^2 + \|\mathbf{v}\|^2 = \|\mathbf{u} + \mathbf{v}\|^2$

Ex 11: Verify that the set of vectors {(1, 0), (0, 1)} is orthogonal with respect to the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = 4\mathbf{u}_1\mathbf{v}_1 + \mathbf{u}_2\mathbf{v}_2$; then, convert it to an orthonormal set by normalizing the vectors.

Ex 12: Let R^3 have the inner product $\langle \mathbf{u}, \mathbf{v} \rangle = \mathbf{u}_1\mathbf{v}_1 + 2\mathbf{u}_2\mathbf{v}_2 + 3\mathbf{u}_3\mathbf{v}_3$. Use the Gram-Schmidt process to transform $\mathbf{u}_1 = (1, 1, 1)$, $\mathbf{u}_2 = (1, 1, 0)$, and $\mathbf{u}_3 = (1, 0, 0)$ into an orthonormal basis.

Best Approximation in Inner Product Spaces

Common problem involves a vector space V whose elements are functions

We can approximate f in V by a function g from a specified subspace W of V

The best approximation to f by functions in W is the orthogonal projection of f onto the subspace W.

$$\hat{p} = proj_{P_2} p = \frac{< p, p_0 >}{< p_0, p_0 >} p_0 + \frac{< p, p_1 >}{< p_1, p_1 >} p_1 + \frac{< p, p_2 >}{< p_2, p_2 >} p_2$$

7.1 Diagonalization of Symmetric Matrices

Definition: A symmetric matrix A is a matrix such that $A^T = A$.

The following matrices are symmetric. $\begin{bmatrix} 7 & -3 \\ -3 & 5 \end{bmatrix}$, $\begin{bmatrix} 1 & 4 & 5 \\ 4 & -3 & 0 \\ 5 & 0 & 7 \end{bmatrix}$, $\begin{bmatrix} d_1 & 0 & 0 & 0 \\ 0 & d_2 & 0 & 0 \\ 0 & 0 & d_3 & 0 \\ 0 & 0 & 0 & d_4 \end{bmatrix}$

Ex 1: What matrices are symmetric?

a. $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$ b. $\begin{bmatrix} 1 & -1 & 0 \\ -1 & 0 & 2 \\ 0 & 2 & 1 \end{bmatrix}$ c. $\begin{bmatrix} a & b \\ c & a \end{bmatrix}$ d. $\begin{bmatrix} a & b & c \\ d & e & f \\ c & f & i \end{bmatrix}$

An example of a type of problem where we use symmetric matrices and wish to diagonalize $2x^2 + 2xy + 2y^2 = 9$ involves identifying the conic represented by $2x^2 + 2xy + 2y^2 = 9$ without examining the graph. This equation can be written $\begin{bmatrix} x & y \end{bmatrix}\begin{bmatrix} 2 & 1 \\ 1 & 2 \end{bmatrix}\begin{bmatrix} x \\ y \end{bmatrix} = 9$. The solution of this requires eigenvalues and eigenvectors in order to rotate this conic to a new set of axes – an orthogonal set. (We will look at this problem in 7.2)

Our next goal is to devise a procedure for orthogonally diagonalizing a symmetric matrix.

Do the following example using methods we already know.

Ex 2: Diagonalize the matrix $A = \begin{bmatrix} 7 & -4 & 4 \\ -4 & 5 & 0 \\ 4 & 0 & 9 \end{bmatrix}$

What is the relationship of the eigenvectors that you found?

Theorem: If A is symmetric, then any two eigenvectors from different eigenspaces are orthogonal.

Definition: Matrix A is orthogonally diagonalizable if there are an orthogonal matrix P and a diagonal matrix D such that $A=PDP^{-1} = PDP^T$

When is orthogonal diagonalization is possible? We are given that $A=PDP^{-1}$ and $P^{-1} = P^T$. Then $A^T = (PDP^T)^T = P^{TT}D^TP^T = PDP^T = A$

Thus the orthogonal diagonalization is possible when A is symmetric.

Theorem: An n x n matrix A is orthogonally diagonalizable if and only if A is a symmetric matrix.

Diagonalization of Symmetric Matrices:

1. Find a basis for each eigenspace of A.
2. Apply the G-S process to each of these bases to obtain an orthonormal basis for each eigenspace.
3. Form the matrix P whose columns are the basis vectors constructed in Step 2: this matrix orthogonally diagonalizes A.

The justification of this procedure is clear: The theorem above ensures that eigenvectors from *different* eigenspaces are orthogonal, while the application of the G-S process ensures that the eigenvectors obtained within the same eigenspace are orthonormal. Thurs, the entire set of eigenvectors obtained by this procedure is orthonormal.

Ex 3: Orthogonally diagonalize the matrix $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$

Ex 4: Assuming that $b \neq 0$, find a matrix that orthogonally diagonalizes $\begin{bmatrix} a & b \\ b & a \end{bmatrix}$

The Spectral Theorem for Symmetric Matrices: If A is a symmetric matrix, then it has the following properties:

a. A has n real eigenvalues, counting multiplicities.
b. The dimension of the eigenspace for each eigenvalue λ equals the multiplicity of λ.
c. The eigenspaces are mutually orthogonal (eigenvectors corresponding to different eigenvalues are orthogonal).
d. A is orthogonally diagonalizable.

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Spectral Decomposition

Suppose $A = PDP^{-1}$, where the columns of P are orthonormal eigenvectors $u_1, \ldots, u_n$ of A and the corresponding eigenvalues $\lambda_1, \ldots, \lambda_n$ are in the diagonal matrix D. Then since $P^{-1} = P^T$,

$$A = PDP^T = \begin{bmatrix} u_1 & \cdots & u_n \end{bmatrix} \begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_n \end{bmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$

$$= \begin{bmatrix} \lambda_1 u_1 & \cdots & \lambda_n u_n \end{bmatrix} \begin{bmatrix} u_1^T \\ \vdots \\ u_n^T \end{bmatrix}$$

Using the column-row expansion of a product, we can write $A = \lambda_1 u_1 u_1^T + \lambda_2 u_2 u_2^T + \cdots + \lambda_n u_n u_n^T$ (2)

This representation of A is called a spectral decomposition of A because it breaks up A into pieces determined by the spectrum (eigenvalues) of A.

Each term in (2) is an n x n matrix of rank 1. For example, every column of $\lambda_1 u_1 u_1^T$ is a multiple of u_1.

Each matrix $u_j u_j^T$ is a projection matrix in the sense that for each $\mathbf{x}$ in R^n, the vector $(u_j u_j^T)\mathbf{x}$ is the orthogonal projection of $\mathbf{x}$ onto the subspace spanned by u_j.

Ex 5: Construct a spectral decomposition of the matrix A that has the orthogonal diagonalization of $A = \begin{bmatrix} 3 & -2 & 4 \\ -2 & 6 & 2 \\ 4 & 2 & 3 \end{bmatrix}$ where

$$P = \begin{bmatrix} u_1 & u_2 & u_3 \end{bmatrix} = \begin{bmatrix} 1/\sqrt{2} & -1/\sqrt{18} & -2/3 \\ 0 & 4/\sqrt{18} & -1/3 \\ 1/\sqrt{2} & 1/\sqrt{18} & 2/3 \end{bmatrix} \text{ and } D = \begin{bmatrix} 7 & 0 & 0 \\ 0 & 7 & 0 \\ 0 & 0 & -2 \end{bmatrix}$$

7.2 Quadratic Forms

Applications include engineering design and optimization, signal processing, potential and kinetic energy, differential geometry, economics and statistics.

Q quadratic form on R^n is a function Q on R^n whose value at a vector $\mathbf{x}$ in R^n is $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ where A is an n x n symmetric matrix. Matrix A is called the matrix of the quadratic form.

Ex 1: Compute quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ for $A = \begin{bmatrix} 4 & 3 & 0 \\ 3 & 2 & 1 \\ 0 & 1 & 1 \end{bmatrix}$

It is important that you be able to find A given the quadratic form.

Summary: The coefficients of x_1^2, x_2^2, . . . go on the diagonal of A. To make A a symmetric matrix, the coefficient of $x_i x_j$ for $i \neq j$ must be split evenly between a_{ij} and a_{ji}.

The x_1x_2 term is called a cross-product term.

If there is no x_1x_2 term, then Q is a diagonal matrix.

Our goal is to eliminate this term in order to determine the rotation.

Ex 2: Find the matrix of the quadratic form.

a. $Q(\mathbf{x}) = 20x_1^2 - 15x_1x_2 - 10x_2^2$

b. $Q(\mathbf{x}) = x_1x_2$

c. $Q(\mathbf{x}) = 4x_1^2 + 4x_2^2 + 4x_3^2 + 4x_4^2 + 3x_1x_2 + 3x_3x_4 - 4x_1x_4 + 4x_2x_3$

Change of Variable in a Quadratic Form

If $\mathbf{x}$ represents a variable vector in R^n, then a change of variable is an equation of the form $\mathbf{x} = P\mathbf{y}$ or $\mathbf{y} = P^{-1}\mathbf{x}$ where P is an invertible matrix and $\mathbf{y}$ is a new variable vector in R^n.

$\mathbf{y}$ is the coordinate vector of $\mathbf{x}$ relative to the basis of R^n determined by the columns of P.

So, to make a change of variable that transforms a quadratic form with a cross product term to a quadratic form with no cross product term, you orthogonally diagonalize the matrix A. The columns of the matrix P form the substitutions needed to transform the equation.

Theorem 4: The Principal Axes Theorem

Let A be an n x n symmetric matrix. Then there is an orthogonal change of variable, $\mathbf{x} = P\mathbf{y}$, that transforms the quadratic form $\mathbf{x}^T A\mathbf{x}$ into a quadratic form $\mathbf{y}^T D\mathbf{y}$ with no cross-product term.

- The columns of P are called the principal axes of the quadratic form $\mathbf{x}^T A\mathbf{x}$
- The vector $\mathbf{y}$ is the coordinate vector of $\mathbf{x}$ relative to the orthonormal basis of R^n given by these principal axes.
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A Geometric View of Principal Axes

Suppose $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ where A is an invertible 2 x 2 symmetric matrix, and let c be a constant.

The set of all $\mathbf{x}$ in R^2 that satisfy $\mathbf{x}^T A\mathbf{x} = c$ corresponds to an ellipse (circle), hyperbola, two intersecting lines, a single point, or no points at all.

If A is diagonal, graph is in standard position.

If A is not diagonal, the graph is not in standard position.

Finding the principal axes (determined by eigenvectors of A) amounts to finding a new coordinate system with respect to which the graph is in standard position.

Classifying Quadratic Forms

Definition: A quadratic form Q is

a) positive definite if $Q(\mathbf{x}) > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
b) negative definite if $Q(\mathbf{x}) < 0$ for all $\mathbf{x} \neq \mathbf{0}$.
c) indefinite if $Q(\mathbf{x})$ assumes both positive and negative values.

Q is positive semidefinite if $Q(\mathbf{x}) \geq 0$ for all $\mathbf{x}$.

Q is negative semidefinite if $Q(\mathbf{x}) \leq 0$ for all $\mathbf{x}$.

Theorem 5 Quadratic Forms and Eigenvalues: Let A be an n x n symmetric matrix. A quadratic form $Q(\mathbf{x}) = \mathbf{x}^T A\mathbf{x}$ is

a) Positive definite iff the eigenvalues of A are all positive.
b) Negative definite iff the eigenvalues of A are all negative.
c) Indefinite iff A has both positive and negative eigenvalues.

In Ex 5:Classify the quadratic form. Then make a change of variable $\mathbf{x} = P\mathbf{y}$ that transforms the quadratic form into one with no cross-product term. Give P and the new quadratic form.

Ex 3: $9x_1^2 - 8x_1x_2 + 3x_2^2$

Note the form of P. P is of the form $\begin{bmatrix} \cos\varphi & -\sin\varphi \\ \sin\varphi & \cos\varphi \end{bmatrix}$ so that we are able to identify the angle of rotation of the conic.

What is the angle of rotation of the conic in Example 3?

Hold Ex4 for the test.

Ex 4: Consider the conic section whose equation is $5x^2 - 4xy + 8y^2 - 36 = 0$. Eliminate the cross-product term and describe the conic. (Hint: We can rewrite this in the form $\mathbf{x}^T A\mathbf{x} - 36 = 0$.) What is the angle of rotation?

In the figure, the vectors $\mathbf{v}_1$ and $\mathbf{v}_2$ are the column vectors of P

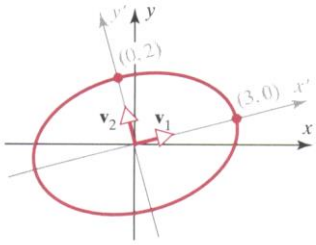


Figure 9.6.4

$$\frac{x'^2}{9} + \frac{y'^2}{4} = 1$$

Identification of the Conic Sections

λ_1, λ_2 both nonzero		Exactly one of λ_1, λ_2 is zero
$\lambda_1 \cdot \lambda_2 > 0$	$\lambda_1 \cdot \lambda_2 < 0$	
Ellipse	Hyperbola	Parabola

Ex 5: Name the conic.

a. $9x^2 - 4xy + 6y^2 - 10x - 20y - 5 = 0$

b. $21x^2 + 6xy + 13y^2 - 114x + 34y + 73 = 0$